Do Domain-Specific Curvature Scales Improve Domain Generalization? A Controlled Re-evaluation of HyperDG

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Abstract

Domain-specific curvature is an appealing inductive bias for domain generalization (DG): if source domains induce different representation geometries, learning and coordinating one curvature scale per domain might improve transfer. We conduct a controlled re-evaluation of this hypothesis using HyperDG, a curvature-inspired training objective on an ImageNet-pretrained ResNet-50. We first state the implemented objective precisely: it rescales tangent features by learned positive curvature magnitudes and regularizes domain means, curvature dispersion, feature-norm dispersion, curvature sensitivity, and local–global loss disagreement. It does not implement a full Lorentz classifier or a Sharpness-Aware Minimization inner loop. We compare the full objective with a matched zero-penalty control, frozen curvature, and no curvature-variance penalty on PACS, VLCS, OfficeHome, and TerraIncognita, using four seeds and all four leave-one-domain-out splits (16 matched runs per cell). The full objective changes mean accuracy by $+0.45$, -0.15, -1.02, and -0.73 percentage points, respectively. None of the 12 planned variant-by-dataset comparisons is significant after Holm correction. Moreover, domain curvature scales remain within 0.001–0.011 of one another even without the variance penalty. These results do not establish equivalence and do not refute hyperbolic representation learning; they show that this specific curvature-scaling objective provides no detectable benefit under the tested pretrained-backbone protocol.

1 Introduction

Domain generalization (DG) seeks a predictor trained on several source domains that remains accurate on an unseen target domain (Wang et al., 2022; Gulrajani & Lopez-Paz, 2020). The setting motivates inductive biases that separate stable task structure from domain-specific variation. One proposal is geometric: hyperbolic spaces efficiently represent hierarchical and multi-scale structure, so domains may benefit from distinct curvature parameters that are subsequently coordinated during training (Chami et al., 2020; Fan et al., 2025; Bi et al., 2025).

That proposal contains two empirical claims that should be tested separately. First, does adding curvature-dependent training terms improve held-out-domain accuracy relative to a matched supervised control? Second, if curvature helps, is a separate parameter for each source domain necessary? The distinction is important. A variance penalty that drives all domain curvatures to the same value may make the per-domain parameterization functionally equivalent to one shared scale. Conversely, a non-significant comparison cannot by itself prove that the parameterizations are equivalent, especially on DG benchmarks with large held-out-domain variation.

We revisit these questions using the released HyperDG implementation. This re-evaluation differs from a conventional method paper in two ways. First, we describe the executed computation rather than the broader geometric motivation. In the controlled ResNet-50 implementation, a positive scalar c_i rescales features from source domain i as $z_i = h_\theta(x)/\sqrt{c_i}$. The objective aligns domain means and penalizes several scalar diagnostics, but it does not perform Lorentz log/exp maps, geodesic classification, parallel transport,

or a SAM inner maximization. Second, we compare variants through the same code path and data partitions, rather than treating literature results produced by different pipelines as controlled baselines.

Our study asks four questions:

1. Does the full curvature-inspired objective improve accuracy over a matched zero-penalty control?
2. Do learning curvature and penalizing curvature variance contribute measurable gains?
3. Do per-domain curvature scales actually diverge during training?
4. Are conclusions stable to component removal and the variance weight?

The answer to the first question is negative within the tested protocol. Across four datasets, the full objective never improves mean accuracy by more than $0.45\,pp$. OfficeHome shows a nominal $-1.02\,pp$ effect, but no comparison survives correction over the 12 planned tests. Frozen-curvature and no-variance variants follow the same pattern. Curvature trajectories explain why the variance term has little leverage: the learned scales remain tightly clustered even when its coefficient is zero.

Our contribution is therefore an empirical boundary, not a new state of the art. We provide (i) an implementation-faithful formulation of the hypothesis, (ii) a 256-run primary controlled study plus 36 diagnostic runs, (iii) paired effect estimates with confidence intervals and multiplicity correction, and (iv) curvature trajectories that identify a mechanism that does *not* materialize. The conclusion is deliberately local: curvature scaling on top of a pretrained Euclidean backbone is not sufficient evidence for the broader claim that hyperbolic geometry improves DG.

2 Related Work

Domain generalization and flatness. DG methods target invariance, robust optimization, data augmentation, or model selection under distribution shift (Gulrajani & Lopez-Paz, 2020). Flat-minima methods such as SAM and SWAD encourage robustness to parameter perturbations or average weights along favorable training regions (Foret et al., 2020; Cha et al., 2021). SFT extends this intuition with cross-domain self-feedback (Li et al., 2025). These methods motivate studying training stability, but parameter-space sharpness and feature-space curvature are different quantities. The implementation evaluated here uses feature-norm and curvature-gradient proxies; we therefore do not describe it as SAM.

Hyperbolic representation learning. Hyperbolic embeddings are effective when data exhibit hierarchy or power-law-like structure (Chami et al., 2020). MERU learns hyperbolic image–text representations from a hyperbolic pretraining objective (Desai et al., 2023). In contrast, our controlled study starts from an ImageNet-pretrained Euclidean encoder and applies a scalar tangent-feature rescaling during fine-tuning. Our conclusions consequently do not transfer to models trained end-to-end with a hyperbolic objective.

Curvature learning and hyperbolic DG. Fan et al. (Fan et al., 2025) study curvature learning for the generalization of hyperbolic neural networks and connect curvature optimization to sharpness-aware bilevel training. Our setting is multi-source DG and our executed parameterization is substantially weaker: a learned scalar rescales features, without a full hyperbolic classifier. Bi et al. (Bi et al., 2025) address fine-grained DG through hyperbolic state-space hallucination, changing the training distribution as well as its geometry. HyperDG uses neither hallucination nor state-space augmentation. The present negative result complements these works by identifying a setting in which curvature-inspired scaling alone does not improve transfer.

3 Method Under Test

3.1 Setting and curvature-scaled features

Let $\mathcal{D}_1, \ldots, \mathcal{D}_N$ be source domains sharing label space $\mathcal{Y}$. An encoder $h_\theta : \mathcal{X} \to \mathbb{R}^d$ and classifier g_ϕ are initialized from the same configuration in every condition. Each source domain has an unconstrained scalar a_i transformed to a positive magnitude

$$c_i = \mathrm{softplus}(a_i) + \epsilon, \qquad \epsilon = 10^{-4}. \tag{1}$$

One may associate $-c_i$ with a nominal negative curvature, but the executed representation is the tangent-space scaling

$$z_i(x) = \frac{h_\theta(x)}{\sqrt{c_i}}. \tag{2}$$

The supervised loss for a source minibatch B_i is $L_i^{\mathrm{sup}} = |B_i|^{-1} \sum_{(x,y)\in B_i} \mathrm{CE}(g_\phi(z_i(x)), y)$, and $L_{\mathrm{sup}} = N^{-1} \sum_i L_i^{\mathrm{sup}}$.

At evaluation, the released implementation predicts with $g_\phi(h_\theta(x))$, because an unseen target has no source-domain curvature index. Thus, the curvature variables act as training-time scales and regularizers; the evaluated classifier is Euclidean. This design choice is central to the scope of our conclusions.

3.2 Auxiliary objective

The full objective contains five auxiliary terms. We name them by their executed computation rather than by legacy configuration labels.

Domain-mean alignment. For minibatch mean $m_i = |B_i|^{-1} \sum_{x\in B_i} z_i(x)$ and size-weighted mean $\bar{m}$, the alignment term is

$$L_{\mathrm{align}} = \frac{1}{N} \sum_{i=1}^{N} \|m_i - \bar{m}\|_2^2. \tag{3}$$

This aligns marginal domain means; it is not a class-prototype or geodesic alignment loss.

Curvature dispersion. The variance-style penalty is implemented as the mean pairwise squared difference

$$L_{\mathrm{var}} = \frac{1}{N^2} \sum_{i,j} (c_i - c_j)^2. \tag{4}$$

It is identically zero in the single-shared-curvature variant.

Norm dispersion proxy. The legacy "sharpness" term is

$$L_{\mathrm{norm}} = \frac{1}{N} \sum_i c_i \, \mathrm{Var}_{x\in B_i}[\|z_i(x)\|_2]\,. \tag{5}$$

It contains no adversarial parameter perturbation and should not be interpreted as SAM or a Hessian-based sharpness estimate.

Local curvature sensitivity. The local term penalizes the magnitude of the supervised-loss derivative with respect to the curvature parameter,

$$L_{\mathrm{local}} = \frac{1}{N} \sum_i \left| \frac{\partial L_i^{\mathrm{sup}}}{\partial a_i} \right|. \tag{6}$$

The derivative is retained in the computation graph, so optimizing this term requires a second-order gradient through a_i.

Local–global consistency. A separate learned scalar c_g defines $z_g(x) = h_\theta(x)/\sqrt{c_g}$. The final term is

$$L_{\text{global}} = \frac{1}{N} \sum_i |L_i^g - \text{stopgrad}(L_i^{\text{sup}})| \,, \tag{7}$$

where L_i^g applies the same classifier to z_g. This penalizes loss disagreement between local and global scales.

The complete training objective is

$$\begin{aligned} L = L_{\text{sup}} + \lambda_{\text{var}} L_{\text{var}} + \lambda_{\text{align}} L_{\text{align}} + \lambda_{\text{norm}} L_{\text{norm}} \\ + \lambda_{\text{local}} L_{\text{local}} + \lambda_{\text{global}} L_{\text{global}}. \end{aligned} \tag{8}$$

The default coefficients are 0.2, 0.5, 0.05, 0.1, and 0.1 in the order shown. Encoder/classifier parameters and curvature parameters use separate Adam optimizers.

3.3 Controlled conditions

We evaluate four conditions through the same implementation:

- ZeroPenalty: all five auxiliary coefficients and the curvature learning rate are zero. This is an ERM-like supervised control inside the HyperDG code path, not a separately implemented DomainBed ERM.
- **Full**: Equation 8 with all default weights and learned per-domain scales.
- **Frozen**: the default auxiliary terms remain active, but curvature learning and L_{var} are disabled.
- **No variance**: per-domain scales remain learnable, but $\lambda_{\text{var}} = 0$.

This design isolates the incremental value of the curvature-inspired terms. It does not support claims about a full Lorentz model or standard ERM outside this code path.

4 Experimental Design

Datasets and protocol. We use PACS (Li et al., 2017), VLCS (Fang et al., 2013), OfficeHome (Venkateswara et al., 2017), and TerraIncognita (Beery et al., 2018). For every dataset, each domain is held out once; the remaining domains are sources. Each environment is split into 80% in-data and 20% out-data using the DomainBed split procedure. The reported outcome is accuracy on the held-out domain's out split at the final training step.

Optimization. All primary conditions use an ImageNet-pretrained ResNet-50 (He et al., 2016) and Adam (Kingma, 2014) with learning rate 5×10^{-5}, batch size 32 per source domain, no weight decay, gradient clipping at 5, and 5,000 updates. Learned curvature parameters use learning rate 10^{-4}. Conditions share seeds, data partitions, backbone, classifier, and training budget.

Replication structure. The primary study contains four seeds and four held-out domains for every condition–dataset cell ($n = 16$ matched observations). It comprises 256 runs. An additional 36 seed-0 runs cover component removal and a PACS variance-weight sweep, for 292 completed controlled runs in total. We report sample means and standard deviations across the 16 seed–split observations. Because repeated seeds within a held-out domain are not independent domain draws, inferential results are treated as descriptive sensitivity analyses rather than population claims over arbitrary domains.

Statistical analysis. For each variant and dataset, we form paired accuracy differences relative to ZeroPenalty using the same seed and held-out domain. We report the mean difference, an unadjusted 95% paired-t interval, and a two-sided paired t-test. Holm's procedure controls family-wise error over the 12 planned (three variants $\times$ four datasets) comparisons. These tests assess whether we detect a difference; failure to reject is not evidence of equivalence. We therefore avoid stronger claims than supported by the interval widths.

Table 1: Held-out accuracy (%) for the controlled study. Each cell is mean ± sample SD over four seeds and four held-out domains ($n = 16$). The matched zero-penalty control shares the HyperDG training path; it is not a separately implemented standard ERM.

Condition	PACS	VLCS	OfficeHome	TerraInc.
ZeroPenalty	84.04 ± 7.10	75.63 ± 13.32	67.05 ± 10.26	43.90 ± 8.49
Full	84.50 ± 7.45	75.47 ± 14.14	66.03 ± 9.98	43.17 ± 4.97
Frozen	84.67 ± 7.42	76.27 ± 13.17	66.13 ± 9.78	44.30 ± 6.74
No variance	84.57 ± 6.92	75.04 ± 14.25	66.03 ± 9.75	43.90 ± 6.94

Table 2: Full objective minus matched control. Intervals are unadjusted 95% paired-t intervals over 16 matched seed–split observations. Holm p is adjusted jointly with the eight frozen/no-variance comparisons shown in Figure 1.

Dataset	Δ (pp)	95% CI	Raw p	Holm p
PACS	+0.45	[−1.06, 1.97]	.534	1.000
VLCS	−0.15	[−1.03, 0.72]	.713	1.000
OfficeHome	−1.02	[−1.79, −0.25]	.013	.153
TerraIncognita	−0.73	[−3.92, 2.46]	.631	1.000

Reproducibility. The repository provides all 292 YAML configurations, the primary per-run accuracy table, aggregation and plotting scripts, and final aggregate tables. Raw checkpoints are not required to reproduce the reported numerical aggregation and are not described as part of the public release.

Superseded preliminary results. An earlier version reported substantially higher single-seed accuracies (for example, 92.2% on PACS) and compared them directly with literature baselines. Those values were not reproducible from the released training path under the controlled protocol and are not used here. The per-run CSV and analysis in this revision supersede the preliminary tables; we do not attribute their gap to the method.

5 Results

5.1 The full objective does not improve the matched control

Table 1 shows no consistent gain from the full objective. Its mean difference relative to ZeroPenalty is positive on PACS and negative on the other three datasets. The largest magnitude is $-1.02\,pp$ on OfficeHome. By comparison, variation across held-out domains produces cell standard deviations of roughly 5–14 points.

OfficeHome is nominally negative before correction, but its Holm-adjusted $p = .153$ (Table 2). Thus, none of the planned comparisons provides multiplicity-adjusted evidence of a difference. The confidence intervals also make the scope clear: PACS and TerraIncognita remain compatible with modest positive or negative effects. The result is "no detected benefit," not proof of exact equality.

5.2 Learning curvature and its variance penalty do not explain a gain

Frozen curvature is never worse than the full objective by more than $0.98\,pp$, and it has higher mean accuracy on three datasets. Removing the variance penalty similarly changes accuracy by less than $0.60\,pp$ relative to full on PACS, VLCS, and OfficeHome; on TerraIncognita it restores the control mean almost exactly. Figure 1 shows that neither variant yields an adjusted difference from ZeroPenalty. These comparisons weaken both candidate mechanisms: improvement does not require learned domain scales, and the variance penalty does not produce an observable accuracy gain.

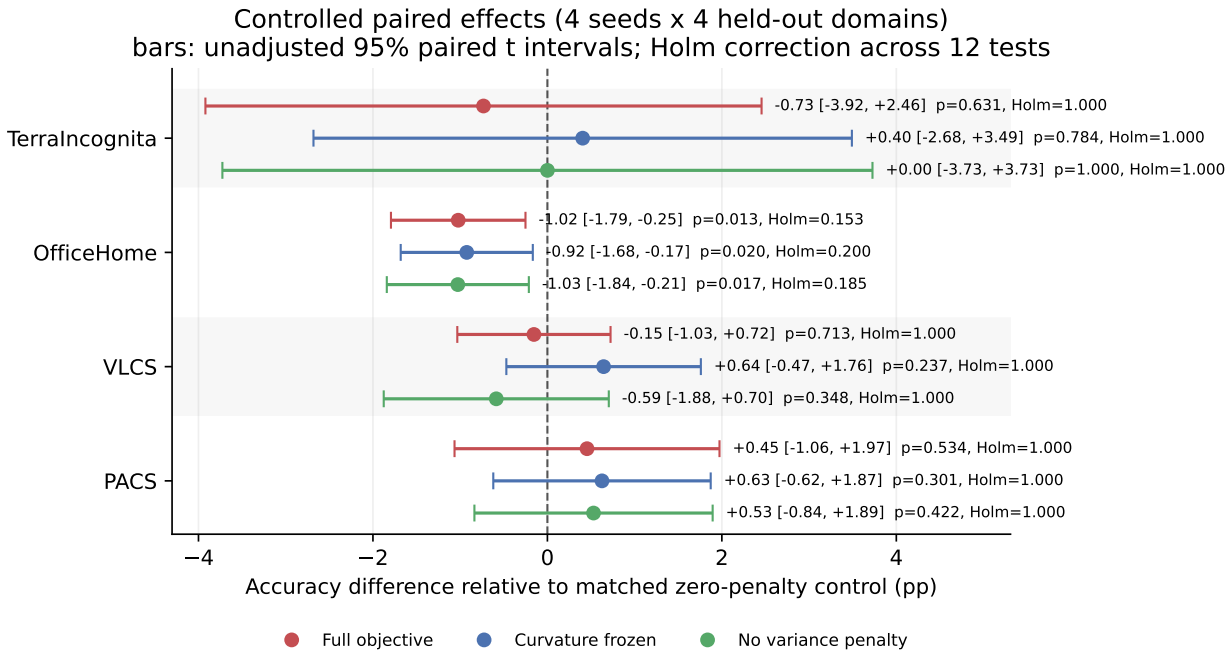


Figure 1: Paired effects relative to ZeroPenalty. Points are mean differences across matched seed–split runs; bars are unadjusted 95% paired-t intervals. Labels report raw and Holm-adjusted p-values. No effect is significant after correcting the 12 planned comparisons.

Table 3: Final within-run curvature range, averaged over four held-out-domain splits. "Max" is the largest range among those splits.

	$\lambda_{\text{var}} = 0$		$\lambda_{\text{var}} = 0.2$	
Dataset	Mean	Max	Mean	Max
PACS	.0041	.0059	.0023	.0037
VLCS	.0040	.0064	.0028	.0049
OfficeHome	.0011	.0016	.0011	.0020
TerraIncognita	.0064	.0106	.0045	.0083

A direct single-shared-scale comparison is available only as a seed-0 diagnostic and is therefore not part of the primary inference. On the three complete four-split datasets, replacing per-domain scales with one shared scale changes mean accuracy by $+0.53\,pp$ (PACS), $-1.28\,pp$ (VLCS), and $-1.23\,pp$ (OfficeHome). The supplementary material reports these values and the incomplete TerraIncognita run transparently. A multi-seed shared-scale study remains necessary for a precise equivalence claim.

5.3 Curvature scales remain close without explicit coordination

Figure 2 directly tests the premise of Equation 4. All scales start near $\text{softplus}(0) + 10^{-4} \approx 0.6932$ and move downward together. Without the variance penalty, the final within-run range is only 0.0011–0.0064 on average across the four datasets (Table 3); even the largest observed split-level range is 0.0106. The penalty narrows the average range on PACS, VLCS, and TerraIncognita, but the absolute change is small, and OfficeHome is unchanged.

This diagnostic explains the weak variance-penalty ablation: the optimization dynamics already keep the scalar parameters close. It does not show that the representations occupy equivalent manifolds, because the implementation does not construct distinct full manifolds in the first place.

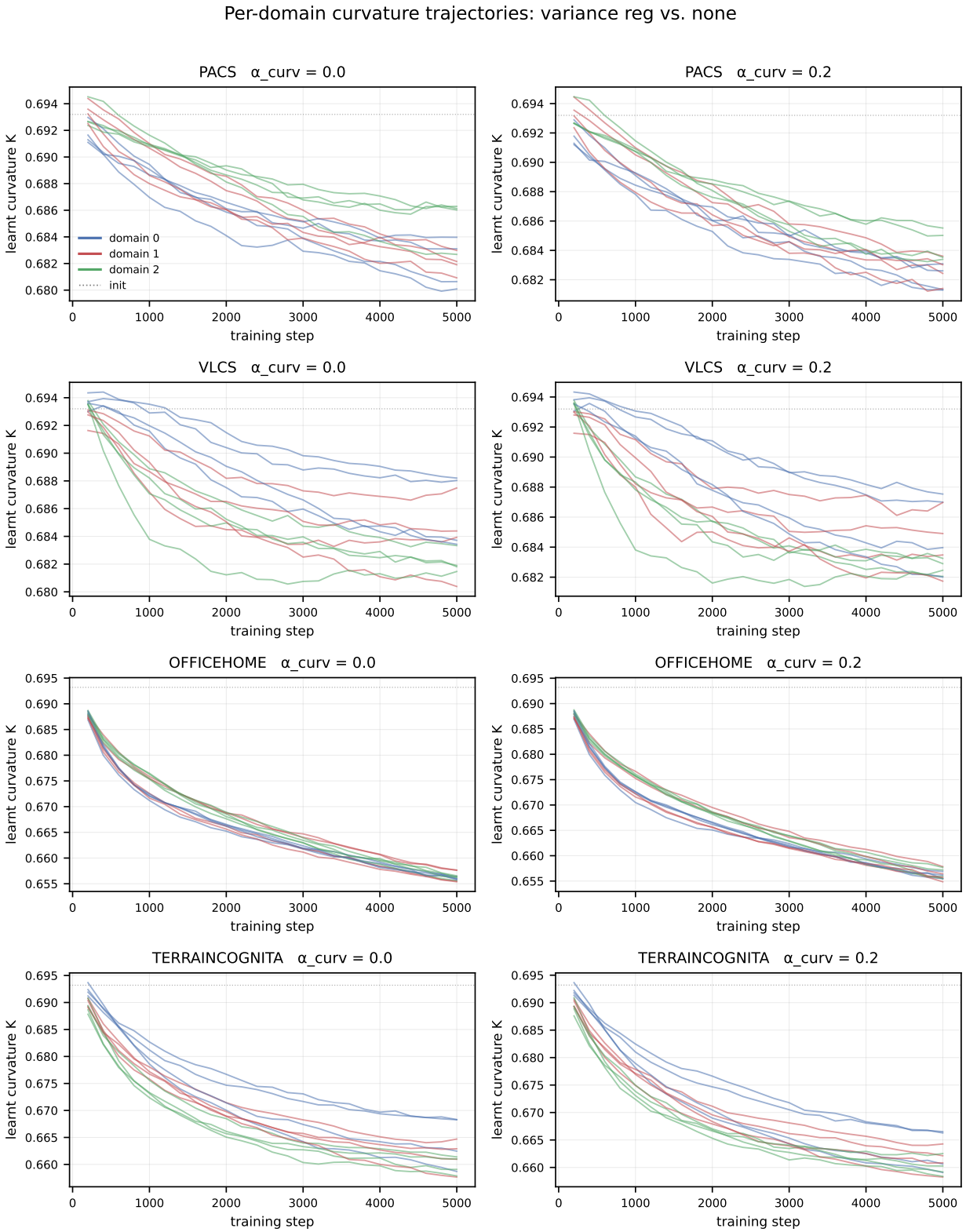


Figure 2: Per-domain curvature magnitudes during training without (left) and with (right) the variance penalty. Each panel contains four held-out-domain splits and three source-domain trajectories per split. The scales move together even when $\lambda_{\text{var}} = 0$.

5.4 Secondary diagnostics are consistent but lower-powered

Seed-0 leave-one-component-out experiments on PACS and OfficeHome do not identify a consistently beneficial auxiliary term. Removing the legacy norm-dispersion "sharpness" proxy increases PACS by 2.30 *pp* but decreases OfficeHome by 0.59 *pp* relative to the full seed-0 result. A four-condition OfficeHome decomposition (zero penalty, proxy terms only, curvature/alignment only, and full) likewise places the full condition last in mean accuracy, but contains only one seed per held-out domain. A PACS sweep over $\lambda_{\text{var}} \in \{0, .05, .2, .5, 2\}$ varies within the large cross-domain dispersion. We treat all three analyses as exploratory; their full tables are in the supplement.

A five-repeat PACS timing benchmark with rotated method order measures 464.27 ± 8.04 ms/update for the full objective and 456.16 ± 26.89 ms/update for genuine DomainBed ERM. The within-repeat relative cost is $1.020 \pm 0.060\times$; peak allocated memory is similar. The computational overhead is therefore modest, although no accuracy benefit is detected in the controlled study.

6 Discussion

What the study establishes. Under a fixed pretrained ResNet-50, 5,000 updates, and the released curvature-scaling implementation, the full auxiliary objective does not provide a detectable improvement over its matched zero-penalty control. Learning a separate scale and penalizing its variance are not supported as mechanisms of improvement. The trajectory evidence is especially informative: the optimization does not generate the curvature divergence that the variance term was designed to prevent.

What it does not establish. The study is not an equivalence test, and several confidence intervals permit non-trivial effects. It also does not test a full Lorentz classifier, a model pretrained with a hyperbolic objective, or a task with an explicit class hierarchy. "Hyperbolic geometry does not help DG" would therefore be an unsupported conclusion. The supported statement is narrower: a particular training-time scalar parameterization and its auxiliary penalties do not help under this controlled protocol.

Why might the intended effect be absent? First, a scalar rescaling is a weak geometric intervention and can be partly absorbed by a linear classifier. Second, the target predictor does not use a learned target curvature, so curvature affects training but not the evaluated representation directly. Third, ImageNet pretraining may dominate a short fine-tuning phase. Finally, PACS, VLCS, and OfficeHome do not expose explicit class hierarchies, the structure for which hyperbolic embeddings have their clearest representational advantage.

A stronger future test. A decisive positive test would train or initialize the encoder with a genuine hyperbolic objective, use geodesic classification consistently at train and test time, evaluate tasks with known hierarchy, and compare one shared versus multiple domain curvatures under a multi-seed matched protocol. MERU-style pretraining (Desai et al., 2023) and hierarchical datasets provide natural starting points. Such a study should also pre-register primary comparisons and use a standard ERM implementation as an external control.

7 Limitations and Broader Impact

Experimental limitations. The primary study covers four image-DG benchmarks and one architecture. It does not include a controlled DomainNet sweep, a true multi-seed shared-curvature condition, or an independently implemented standard ERM baseline. The 16 seed–split observations per cell reuse four held-out domains, limiting population-level inference. Hyperparameter and component analyses are seed-0 diagnostics. We report final-step accuracy rather than a separately tuned model-selection rule. These choices bound the claim to the configurations actually tested.

Implementation limitations. The evaluated code is curvature-inspired rather than a complete hyperbolic network. Post-hoc Poincare visualizations in the supplement apply PCA and a fixed-curvature exponential

map after training; they are qualitative views, not evidence that training occurred on the Poincare disk. The current MERU and Camelyon17 experiments lack the controls required for conclusions and are not used as confirmatory evidence.

Broader impact. DG is often motivated by deployment across sites, devices, or populations. Negative evidence can reduce the risk of adopting additional complexity without measurable benefit, but it should not be interpreted as a guarantee of safety or robustness. In particular, the exploratory Camelyon17 setting is medical; no clinical claim follows from these experiments. Deployment in healthcare or other high-stakes settings requires representative external validation, subgroup analysis, calibration, and monitoring beyond average held-out-domain accuracy.

8 Conclusion

We re-evaluated whether domain-specific curvature scales and their auxiliary regularizers improve domain generalization. In 256 matched primary runs, the full objective produced no multiplicity-adjusted accuracy difference from a zero-penalty control on PACS, VLCS, OfficeHome, or TerraIncognita. Frozen and no-variance variants led to the same conclusion, while curvature trajectories showed that the domain scales remain close even without explicit coordination. The result narrows, rather than rejects, the hyperbolic-DG hypothesis: the tested scalar training-time parameterization is insufficient, and stronger claims require a genuinely hyperbolic model, a standard external control, and a multi-seed shared-curvature comparison.

References

Sara Beery, Grant Van Horn, and Pietro Perona. Recognition in terra incognita. In *Proceedings of the European conference on computer vision (ECCV)*, pp. 456–473, 2018.

Qi Bi et al. Learning fine-grained domain generalization via hyperbolic state space hallucination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 2025.

Junbum Cha, Sanghyuk Chun, Kyungjae Lee, Han-Cheol Cho, Seunghyun Park, Yunsung Lee, and Sungrae Park. Swad: Domain generalization by seeking flat minima. *Advances in Neural Information Processing Systems*, 34:22405–22418, 2021.

Ines Chami, Albert Gu, Vaggos Chatziafratis, and Christopher Ré. From trees to continuous embeddings and back: Hyperbolic hierarchical clustering. In *NeurIPS*, 2020.

Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, and Ramakrishna Vedantam. Hyperbolic image-text representations. In *International Conference on Machine Learning*, 2023.

Xiaomeng Fan et al. Curvature learning for generalization of hyperbolic neural networks. *International Journal of Computer Vision*, 133(12):8489–8525, 2025.

Chen Fang, Ye Xu, and Daniel N Rockmore. Unbiased metric learning: On the utilization of multiple datasets and web images for softening bias. In *Proceedings of the IEEE international conference on computer vision*, pp. 1657–1664, 2013.

Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. *arXiv preprint arXiv:2010.01412*, 2020.

Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. *arXiv preprint arXiv:2007.01434*, 2020.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.

Diederik P Kingma. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

Aodi Li, Liansheng Zhuang, Xiao Long, Minghong Yao, and Shafei Wang. Seeking consistent flat minima for better domain generalization via refining loss landscapes. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 15349–15359, 2025.

Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M Hospedales. Deeper, broader and artier domain generalization. In *Proceedings of the IEEE international conference on computer vision*, pp. 5542–5550, 2017.

Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 5018–5027, 2017.

Jindong Wang, Cuiling Lan, Chang Liu, Yidong Ouyang, Tao Qin, Wang Lu, Yiqiang Chen, Wenjun Zeng, and Philip S Yu. Generalizing to unseen domains: A survey on domain generalization. *IEEE transactions on knowledge and data engineering*, 35(8):8052–8072, 2022.